Formal Lab Report “Unknown A”

Hannah Sommers

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Introduction

This report seeks to elucidate the identity of an unidentified compound, denoted as A, by scrutinizing three distinct graphs: FTIR GCMS, and NMR. Each graph provides unique insights into the characteristics of the unknown compound. Proficiency in deciphering these three graphs is an essential and practical skill for determining the compound's name and molecular formula.

Body

FTIR

Fourier-transform infrared spectroscopy (FTIR) uses the number of radiation wavelengths absorbed through a specified object. It provides whether or not a functional group is present within the compound. Each functional group frequency range is distinguishable from one to another. This allows the ability to determine the different composition and chemical bonds of the compound.

Vibrational spectroscopy is where covalent bond vibrations are comparable to springs on atomic nuclei. When the photons during IR radiation are absorbed in the right amount it can either bend or stretch the covalent bonds. An absorption band occurs here when a molecular vibration happens within the molecule. It can only be quantized when the infrared energy is the same frequency as the molecular vibration.

Three types of molecular vibration exist here; stretching, symmetric stretching, and bending vibrations. Stretching vibrations occur as the distance increases or decreases rhythmically while the atoms are still aligned in the bond axis. Symmetric vibrations happen when atoms stretch in and out simultaneously. Lastly, bending vibrations are how they sound, the position of said atoms changes relative to their bond axis.

Using Fourier-transform infrared spectroscopy for more than ten atoms can become severely complex. The absolute number of observed absorption bands differs from the possible combinations of fundamental vibration. An overtone band, an intense absorption peak, can and

will still be present within these molecules. It appears from the change present of one or more vibrational energy levels during the absorption of infrared spectroscopy. These can be examples of C=O, C=N, and C=C.

Understanding the frequencies and types of peaks will ensure the accuracy of identifying functional groups within the compound. Fourier-transform infrared spectroscopy is never used alone since it only identifies the functional groups within the molecule. Fourier-transform infrared spectroscopy is always paired with H NMR to help predict the complete compound.

GCMS

Gas Chromatography-Mass Spectrum (GCMS) is a method that vaporizes compounds that are usually liquid or solid. The molecules are ionized instantly by the electron impact leaving the resulting ions to be accelerated and passed through a mass analyzer. This determines the weight of each molecule within the compound allowing for a more accurate picture of the composition of the compound. It is limited to compounds with limited vapor pressures.

The Rule of Thirteen is primarily used during this analytical method. The Rule of Thirteen, when used, can generate the chemical formula of a hydrocarbon when using the mass divided by the charge number of the molecular ion. This is then further divided by thirteen to obtain the number of carbons present within the compound. The nitrogen rule is also used to further explain that the compound that has an even mass divided by the charge number of the molecular ion may either carry no nitrogens or an even amount of them. To further explain if mass divided by the charge number of the molecular ion is odd then the compound must carry an odd amount of nitrogens.

The M + peak depends on the isotopic abundances of atoms within a select molecule and the number of each different atom. Each isotope comparable to carbon, hydrogen, oxygen,

and nitrogen, makes up a small contribution to the M + peaks. If the present molecular ion is low in relative abundance, the precision of the M + peak is insufficient for ratios, making the test unreliable. While this is a factor to consider M + peak unreliable, it is still highly valuable for quantitative data that is presented for halogens and other molecules.

Isotope abundance and probability help determine the intensity and mass of particular atoms when duplicates of it are present. If said molecule is not present within a library, fragmentation is used as a hint source to the possibility of what compound it is. It is used alongside ions or free radicals to determine the possible location and destination at which an electron attaches itself. Carbocations give off the location of electrons due to the positive charge.

NMR

Nuclear magnetic resonance (NMR) spectroscopy is one method used to determine and analyze the structure of compounds in chemistry. It helps visualize where and how atoms are connected and bonded together. It also utilizes location and specialized ranges for functional group labeling.

H NMR is a type mainly used in this class and laboratory. It shows how many hydrogens are connected to a specified bond in a compound. It can inform the scientist of the number of different protons, the number of protons per signal, patterns signaling spin-spin coupling, and the signals are located at the horizontal axis of the graph. A specified amount of hydrogens and their location can signal if it is an ethyl, isobornyl, or tert-butyl group.

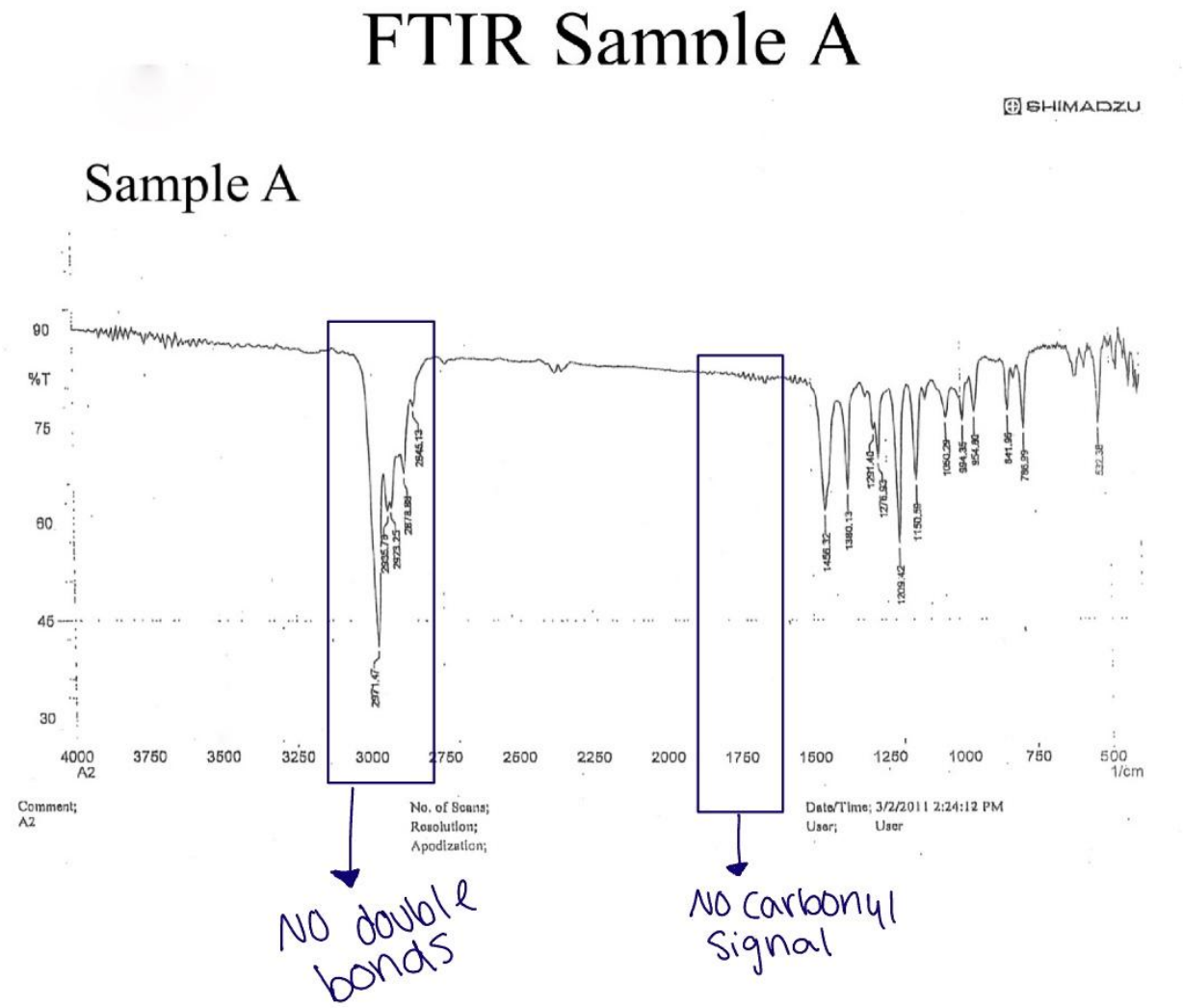
The chemical shift based on the hydrogen nucleus is highly influenced by the electron density that surrounds it. While it undergoes an applied magnetic force the circulation electrons that are in the spherical electron cloud influence a magnetic that counteracts the applied

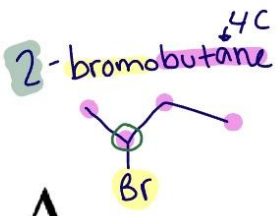
magnetic force. This is to shield the hydrogen nucleus from the applied magnetic force giving off a local diamagnetic shield.

The diamagnetic shield protects said hydrogen nucleus from induction possibly happening. But, if the electron density is decreased the induced magnetic field will decrease as well. This introduces deshielding making the proton more susceptible to the applied magnetic force. Electron-withdrawing substituents can also decrease the deshielding effect the further it is from a halogen.

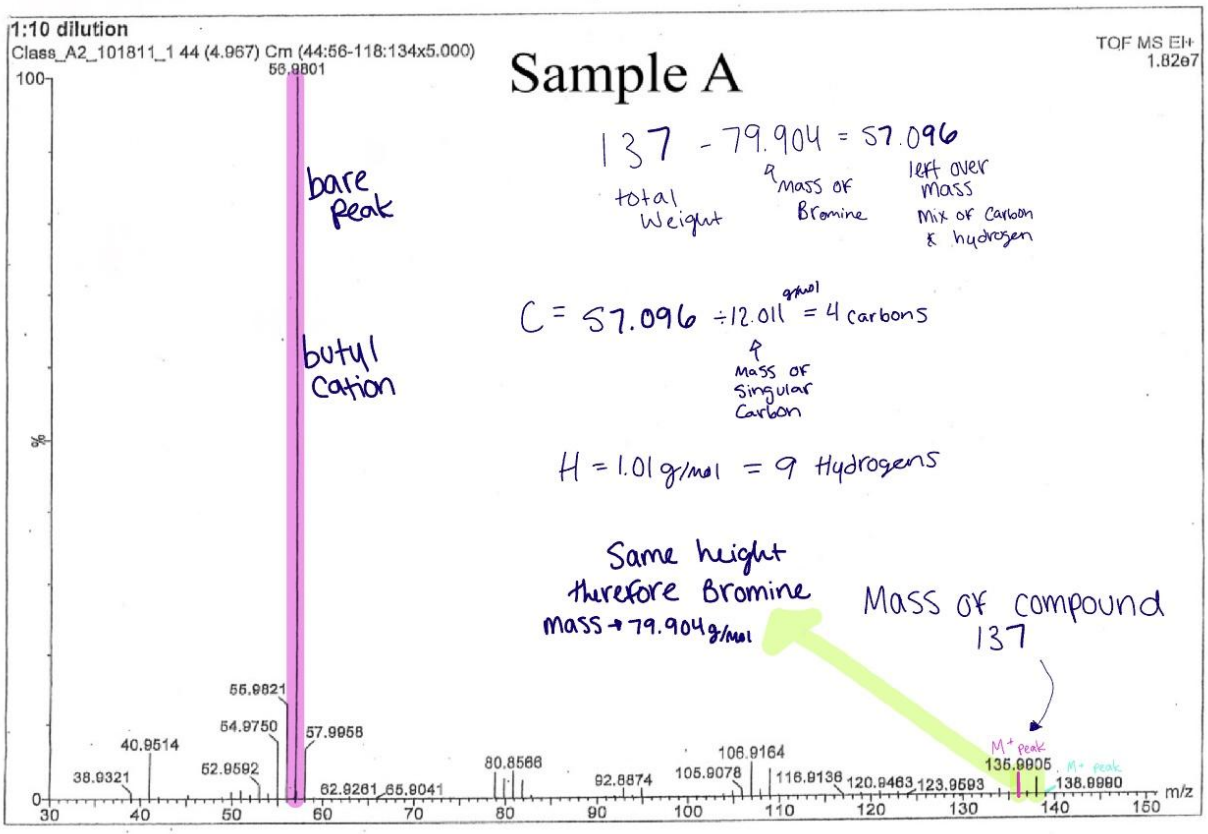
The most important aspect of this specific method is understanding the N + 1 rule. The N + 1 rule is the number of hydrogens plus one to give the accurate counting of said peaks. If the amount of hydrogens is greater than N + 1, it is not correct and the peaks will be over the amount needed. N specifically in this case is the number equivalent to the adjacent protons on a given carbon atom. Being able to correctly and efficiently able to identify the number of peaks to the number of hydrogens is a very good and effective way to apply the knowledge learned.

Data and Analysis

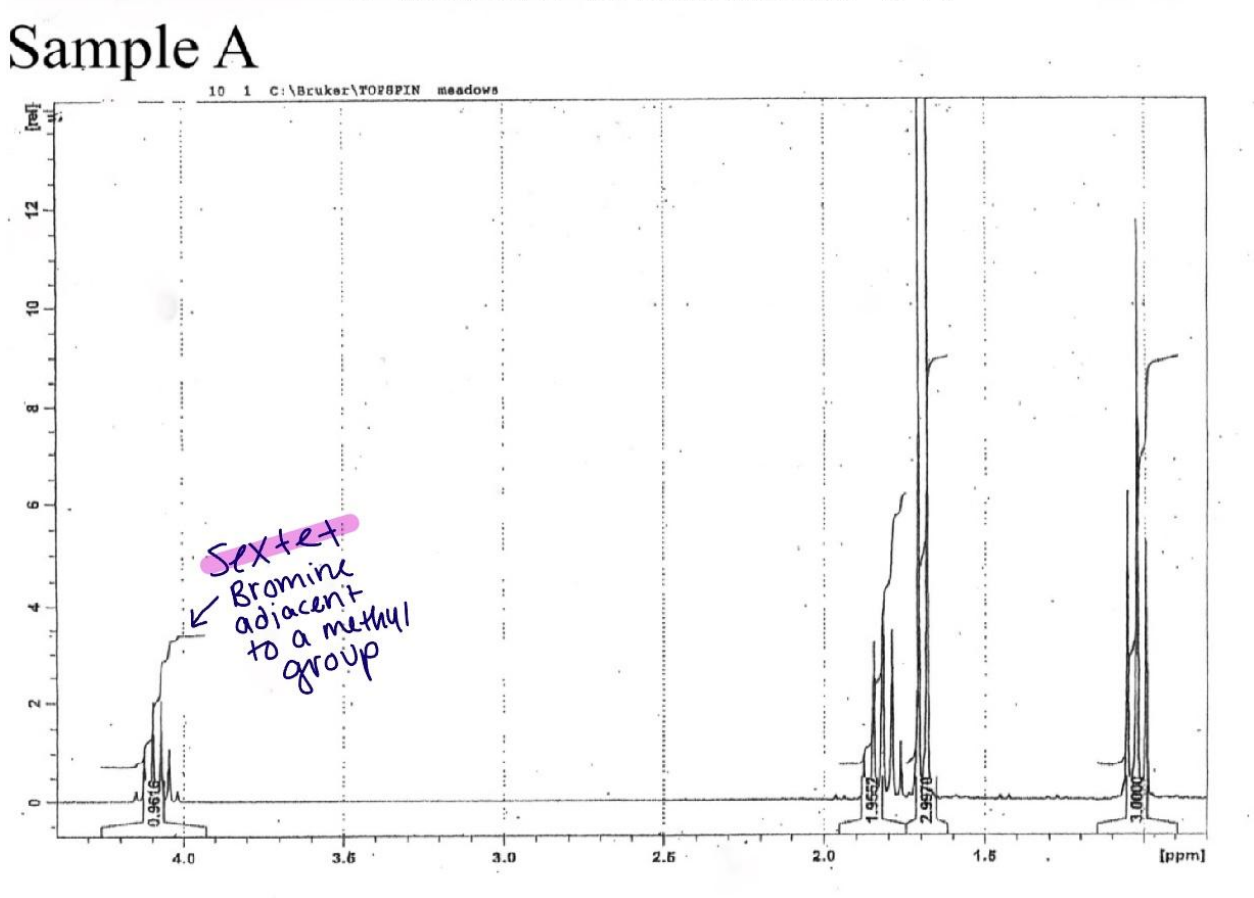




GCMS Sample A



NMR Sample A

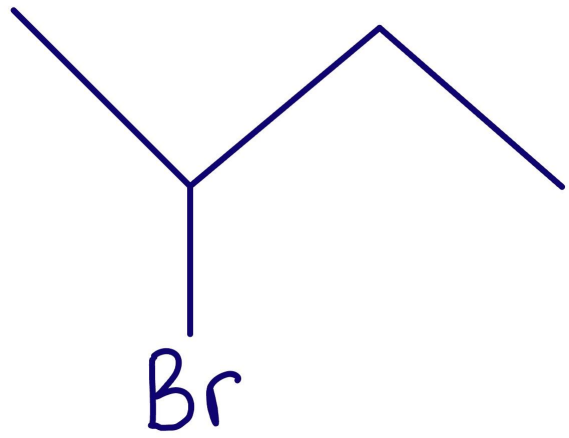


Conclusion

The findings derived from the analysis of NMR, FTIR, and GCMS suggest that the unidentified compound corresponds to 2-bromobutane. In the Fourier-transform infrared spectroscopy (FTIR) graph, no peaks were observed between 1600-1900 cm-1 signifying the absence of carbonyl groups in the compound. In the region near 3000 cm-1 was observed that no double bonds are present within the compound. There is a peak at 2971.47 cm-1 which indicates a stretching of a C-H bond. Multiple possibilities were eliminated, but more data was needed to be analyzed to fully conclude what the compound is. This is where analyzing the GCMS and NMR graph provides more accuracy in identifying the unknown compound.

After analyzing the Gas Chromatography-Mass Spectrum (GCMS) spectrum, the molecular weight of the compound was determined. The tallest peaks farthest to the right are the M+ peaks; these peaks are 136 m/z and 138 m/z shown by $[C_4H_9{}^{79}Br]^+$ and $[C_4H_9{}^{81}Br]^+$ respectively. This suggests the weight of the compound shown is 137 g/mol. The M+2 peak was measured to be at 139 m/z. The M+ peaks are seen to be around the same height spaced by a single m/z indicating a bromine is present in the compound. The molecular mass of bromine is 79.904 g/mol per any given periodic table which comes from both isotopes 79-Br and 81-Br. The molecular weight of 79-Br is subtracted from the total molecular weight of the $[C_4H_9{}^{79}Br]^+$, leaving 57 m/z from the remaining mixture of carbons and hydrogens. Carbon has a molecular mass equivalent to 12.00. The remaining mass, 57 m/z, is divided by 12.00 to equal four carbons. The total mass of the carbons equals about 48 g/mol leaving 9.00 m/z of hydrogens remaining. The molecular mass of hydrogen is about 1 m/z for 1-H, leaving nine hydrogens remaining in the compound. This suggests the chemical formula of C4H9Br for the unknown molecule. With this formula, the degrees of unsaturation can be determined to be 0 which supports the IR data by suggesting there are no unsaturated bonds.

Nuclear magnetic resonance (NMR), is then used to determine the structure of the compound. The four signals on the graph inundated the four different hydrogen groups within the compound. The sextet at 4.1 ppm is the hydrogen adjacent to the bromine. It also presents five neighboring hydrogens based on their multiplicity and integration out to only one hydrogen. From analyzing this graph, the final structure of the compound can be determined and named 2-bromobutane.



References

Ruychev, Veselin. *CHEM 2081 ORGANIC CHEMISTRY II LAB MANUAL 2023 EDITION Department of Chemistry| Auburn University*. Department of Chemistry| Auburn University, 2023. (pg. 89-174)